E -> R LIM TAU_{>=-N}E IS AN ISOMORPHISM UNDER PER-POINT VANISHING

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Truncation triangles. Derived Categories, Remark derived-remark-truncation-distinguished-triangle. Distinguished triangles H^{-n-1}(E)[n+1] -> K_{n+1} -> K_n -> H^{-n-1}(E)[n+2].
- Stalkwise injectivity criterion. Cohomology of Sheaves, Lemma cohomology-lemma-cohomology-derived-limit-injective. Injectivity of H^m(R lim K_n)_x -> H^m(K_{n(x)})_x from the two per-point assumptions.

Source reference. Original reference 0H72: sources/stacks-chow/tags.txt, source lines 21442–21442. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D61: sources/stacks-chow/tags.txt, source lines 16506–16506. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 08J5: sources/stacks-chow/tags.txt, source lines 10838–10838. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $E \in D(\mathcal{O}_X)$. Assume that for every $x \in X$ there exist a function $p(x, -) : \mathbf{Z} \to \mathbf{Z}$ and a fundamental system $\mathfrak{U}_x$ of open neighbourhoods of x such that*

$$H^p(U, H^{m-p}(E)) = 0 \text{ for } U \in \mathfrak{U}_x \text{ and } p > p(x, m)$$

Then the map $E \to R\lim \tau_{\geq -n}E$ of Derived Categories, Remark 0H72 is an isomorphism in $D(\mathcal{O}_X)$.

Proof. Set $K_n = \tau_{\geq -n}E$ and $K = R\lim K_n$. The canonical map $E \to K$ comes from the canonical maps $E \to K_n = \tau_{\geq -n}E$. We have to show that $E \to K$ induces an isomorphism $H^m(E) \to H^m(K)$ of cohomology sheaves. In the rest of the proof we fix m. If $n \geq -m$, then the map $E \to \tau_{\geq -n}E = K_n$ induces an isomorphism $H^m(E) \to H^m(K_n)$. To finish the proof it suffices to show that for every $x \in X$ there exists an integer $n(x) \geq -m$ such that the map $H^m(K)_x \to H^m(K_{n(x)})_x$ is injective. Namely, then the composition

$$H^m(E)_x \to H^m(K)_x \to H^m(K_{n(x)})_x$$

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is a bijection and the second arrow is injective, hence the first arrow is bijective. Set

$$n(x) = 1 + \max\{-m, p(x, m-1) - m, -1 + p(x, m) - m, -2 + p(x, m+1) - m\}.$$

so that in any case $n(x) \geq -m$. Claim: the maps

$$H^{m-1}(U, K_{n+1}) \to H^{m-1}(U, K_n) \quad \text{and} \quad H^m(U, K_{n+1}) \to H^m(U, K_n)$$

are isomorphisms for $n \geq n(x)$ and $U \in \mathfrak{U}_x$. The claim implies conditions (1) and (2) of Lemma 0D61 are satisfied and hence implies the desired injectivity. Recall (Derived Categories, Remark 08J5) that we have distinguished triangles

$$H^{-n-1}(E)[n+1] \to K_{n+1} \to K_n \to H^{-n-1}(E)[n+2]$$

Looking at the associated long exact cohomology sequence the claim follows if

$$H^{m+n}(U, H^{-n-1}(E)), \quad H^{m+n+1}(U, H^{-n-1}(E)), \quad H^{m+n+2}(U, H^{-n-1}(E))$$

are zero for $n \geq n(x)$ and $U \in \mathfrak{U}_x$. This follows from our choice of $n(x)$ and the assumption in the lemma. $\square$